

Rahul Rawat

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PROFILE

Data Science Master's student at SRH Heidelberg with hands on experience building and shipping AI based applications from end to end, from a modular retrieval augmented generation system with agentic routing to a fully automated cloud data pipeline. Comfortable across Python, modern AI APIs, large language model prompting, retrieval augmented generation, embeddings, and vector databases. Curious by nature, with a track record of benchmarking models, evaluating new frameworks, and turning prototypes into deployed tools. Looking to bring that AI engineering mindset to valantic's SAP AI team while learning SAP Cloud Native in practice.

EDUCATION

M.Sc. Data Science and Analytics Apr 2025 to Present
SRH University of Applied Sciences Heidelberg

Bachelor's Degree in Computer Science 2019 to 2023
GL Bajaj Institute of Technology and Management

EXPERIENCE

eRay GmbH Oct 2025 to Mar 2026
Data Scientist

- Built a complete recursive forecasting pipeline covering four water quality indicators for a German lake: chlorophyll a, turbidity, pH, and dissolved oxygen, across a six month academic and industry collaboration.
- Benchmarked six models head to head, including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, then used CatBoost multi quantile regression to deliver asymmetric 80 percent prediction intervals.
- Enforced strict rules against data leakage, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not predictable without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast curve to stop tree models flatlining during recursive prediction.
- Wrapped the pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade.

PERSONAL PROJECTS

Hybrid RAG Orchestrator: Python, LangChain, Llama 3.1 8b, ChromaDB, HuggingFace, Streamlit

- Designed a modular retrieval augmented generation system using Llama 3.1 8b through Groq for high speed inference, with LangChain handling orchestration across the whole pipeline.
- Engineered a custom router that dynamically classifies user intent into three execution paths: local knowledge retrieval from PDFs and a vector store, external web search through DuckDuckGo, or direct conversational logic, the same agentic routing pattern behind real client AI prototypes.
- Implemented ChromaDB with local persistence for document storage, using HuggingFace MiniLM embeddings for semantic search.
- Built a stateful memory agent that holds conversational context across multiple turns and is wired directly into the inference pipeline, then shipped the whole system behind a Streamlit interface with full ownership from design through deployment.

Real Time Flight Tracking Data Pipeline: Python, PySpark, BigQuery, dbt, Apache Airflow, Google Cloud Platform, Tableau

- Set up real time data collection from the OpenSky Network API, polling live flight positions every 30 seconds and enriching each one against airport, aircraft, and weather data from four separate sources.
- Cleaned raw data with PySpark on Google Cloud and shaped it into analysis ready tables with dbt, computing each aircraft's nearest airport along the way.
- Orchestrated the whole system with Apache Airflow so the batch and real time layers refresh automatically every 15 minutes.
- Surfaced findings in Tableau using Python statistics through TabPy, showing that air traffic dropped 4.4 times over in heavy rain and clustered around hubs like Frankfurt and Munich.

Movie Analytics and ML Pipeline, a Cloud Native Data Platform: Google Cloud Platform, BigQuery, Cloud Run, Cloud Scheduler, Python, SQL, Looker Studio

- Built a complete batch pipeline that pulls movie data from a public API into a GCS data lake, processes it through Bronze to Silver to Gold medallion architecture in BigQuery, and runs fully automated on a Cloud Scheduler trigger with no manual steps.
- Engineered the Silver layer for trustworthy analytics, covering schema enforcement, safe type casting, deduplication through window functions, and genre normalization into a relational model.
- Trained a BigQuery ML classifier to predict whether a film will be a hit before release, deliberately splitting features into two tables to prevent data leakage by using only pre release signals.
- Secured the system with a least privilege service account and Secret Manager, and version controlled all code, SQL, and schemas in a reproducible GitHub repo.

RESEARCH AND THESIS

Bachelor Thesis: Diabetes Prediction Using Machine Learning

Python, Scikit Learn, Pandas, Seaborn, Google Colab. Written up as an IEEE style paper.

- Built a complete machine learning pipeline comparing six classifiers on a clinical dataset of 768 patient records, using 10 fold cross validation and per model confusion matrices.
- Caught biologically impossible zero values the original authors had overlooked, then applied IQR based outlier removal and proper imputation.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure.
- Wrote up findings as an IEEE style paper, including an honest section on limitations and what to do differently.

LANGUAGES

English: fluent, professional working proficiency

German: A2.2, completed through VHS